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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} &= \frac{1 - 1}{0} = \frac{0}{0} \\ \Rightarrow \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(e^x - e^{-x})}{\frac{d}{dx}x} = \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{1} = 2\end{aligned}$$

References